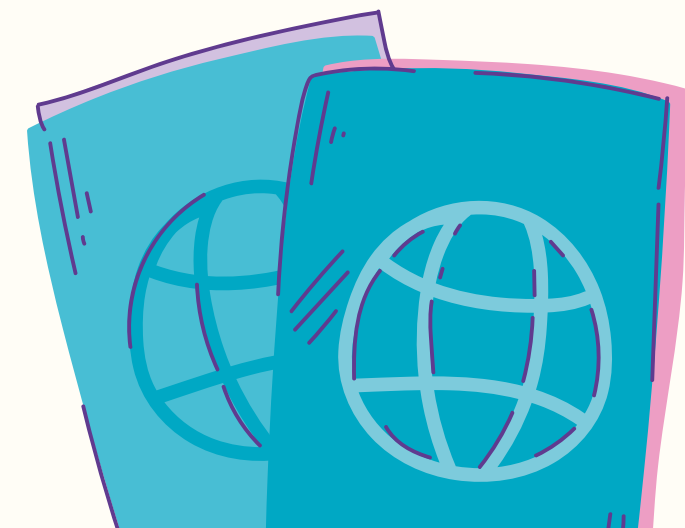
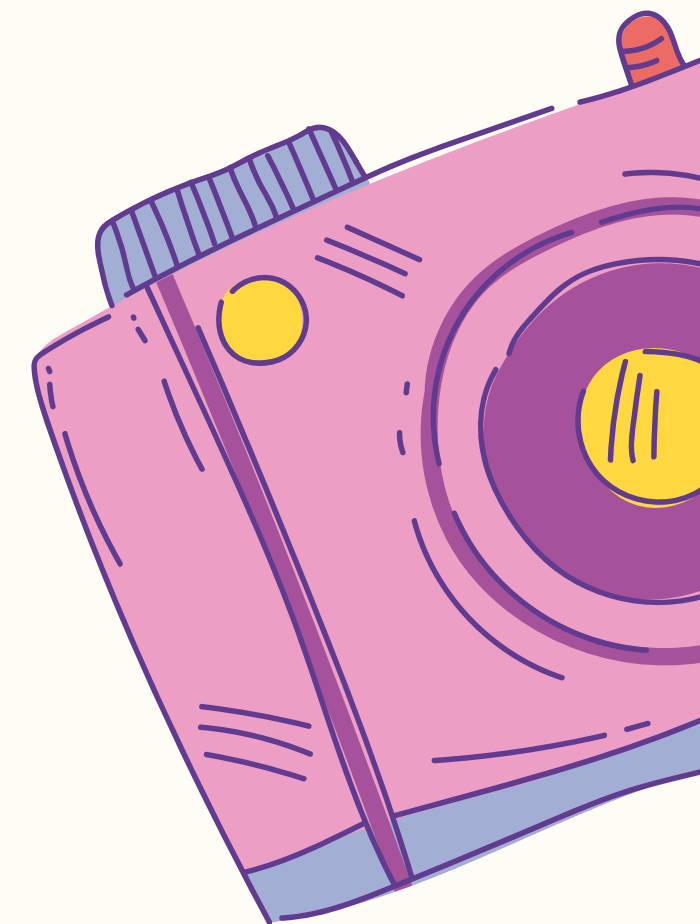
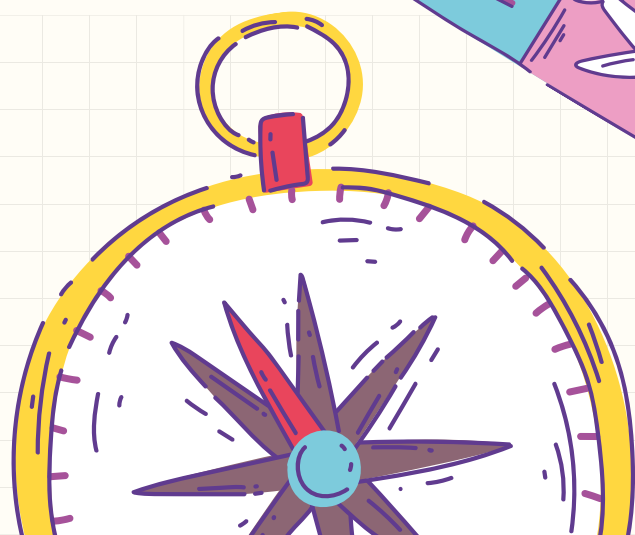
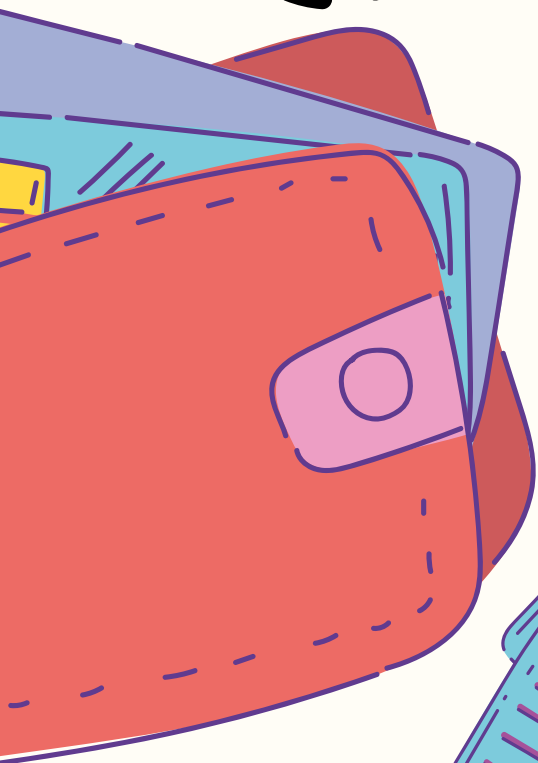


XploreQuest

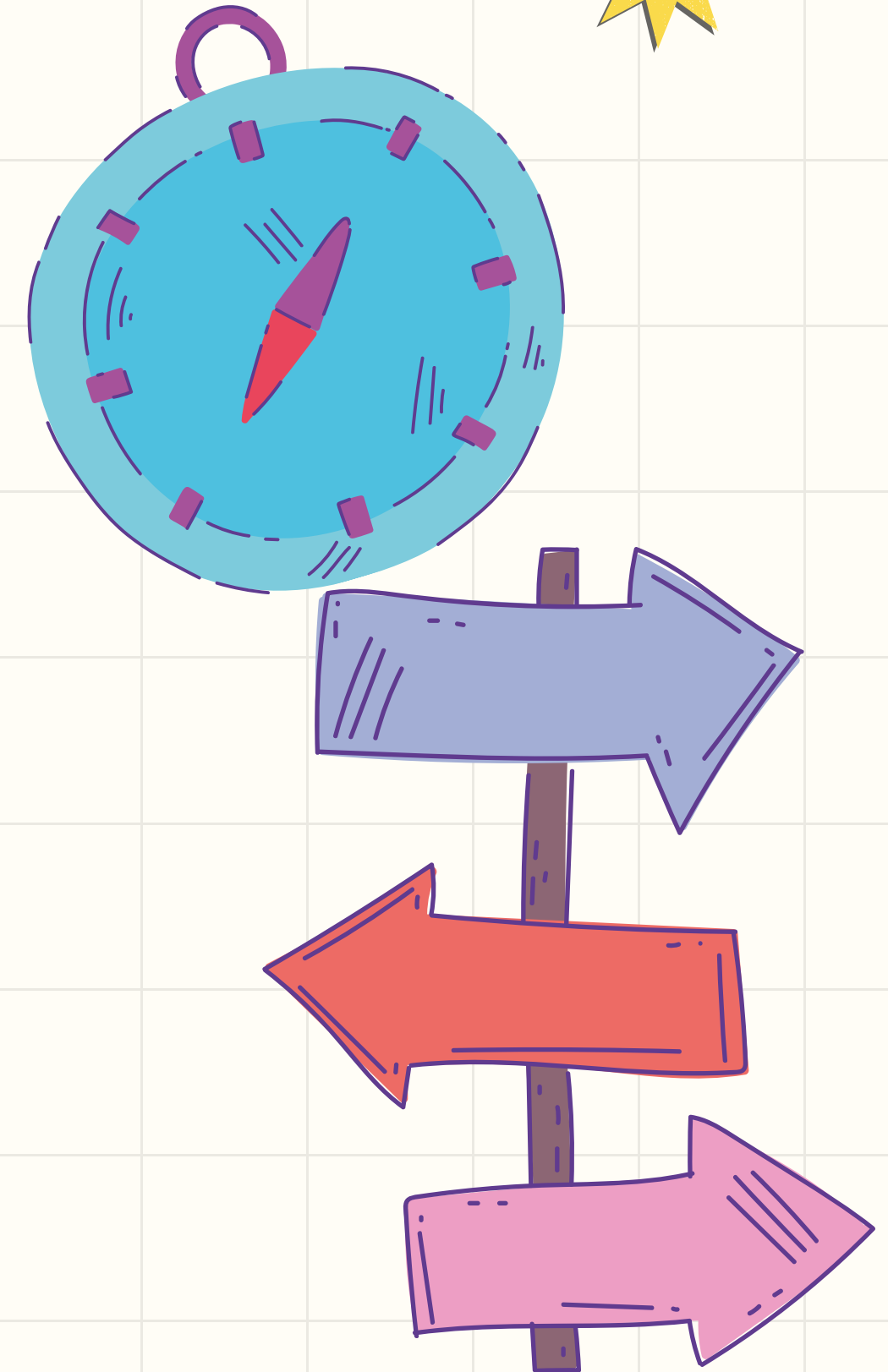
A Secure Cross-Platform Mobile Exploration Application

Presented by: Muhammad Syah Bin Razak
AM2412018244
SV: Madam Nur Afra Binti Taidin



Introduction/ Project Background

- ★ Digital Evolution of Adventure Races (Exploraces): Transitioning outdoor adventure races (Exploraces) from slow, paper check-ins to automated mobile geofencing.
- ★ Client Operational Needs: Partnering with KL Event Crew to manage registrations, eliminate checkpoint bottlenecks, and run live leaderboards.
- ★ Cheating & Verification Vulnerabilities: Addressing common exploits in location-based applications, specifically GPS spoofing and QR screenshot sharing.
- ★ XploreQuest Solution: A secure Expo/Firebase app implementing dynamic HMAC-signed QR codes, GPS velocity checks, and offline SQLite synchronization.



Problem Statement

- ✦ Lack of Security in the Checkpoint Validation Process
- ✦ Insecure Caching of Session Credentials on Mobile Devices
- ✦ Absence of Offline Support in Low-Connectivity Environments

Project Objectives

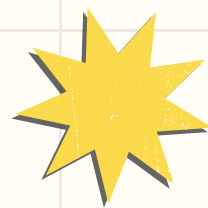
- ✦ To develop a secure checkpoint validation system
- ✦ To design a secure local storage architecture with token encryption
- ✦ To implement an offline synchronization queue for checkpoint scans



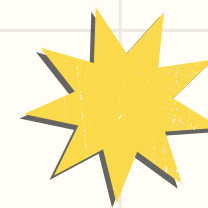


Project Scope

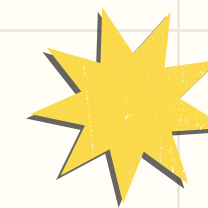
XploreQuest project will develop a secure mobile exploration application that addresses critical vulnerabilities in KL Event Crew's current manual checkpoint processes.

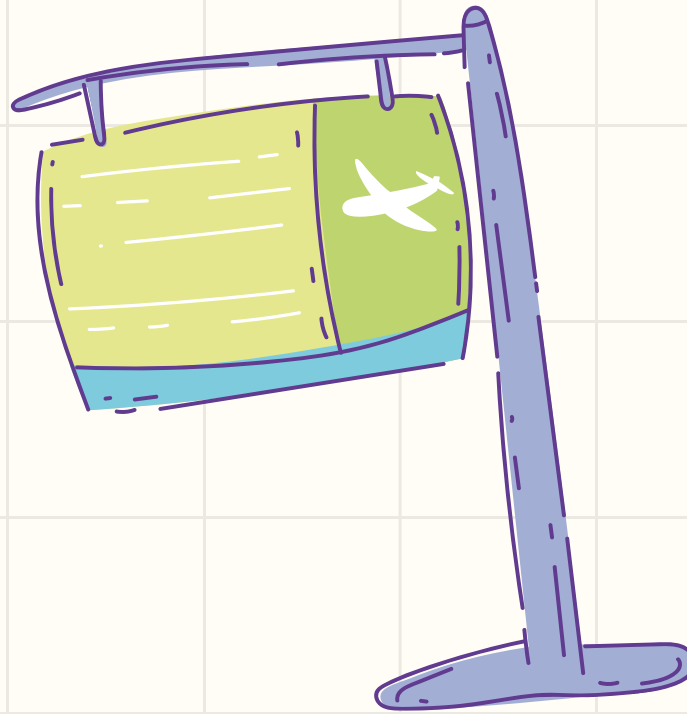


Implementation focus on: preventing coordinate and check-in cheating, securing local session credentials, maintaining offline first synchronization during network dropouts



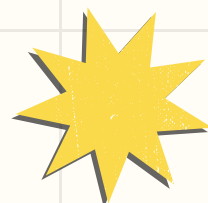
Involve: designing user interfaces for geofenced race maps, developing role-based access controls, implementing dynamic QR code verification and GPS telemetry check algorithms



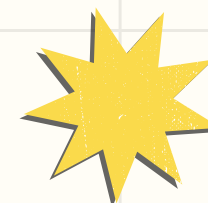


Product Scope

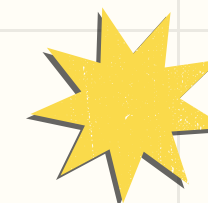
Participant mobile application: Can register, join a team, view geofenced race map and clues, scan checkpoint QR codes, and view active team progress



Checkpoint marshal view: Can verify team physical arrival, update challenge status, and generate dynamic team-specific QR codes



Administrative web platform: Can create events, configure boundaries and checkpoints, stagger team starting points, and monitor live leaderboards

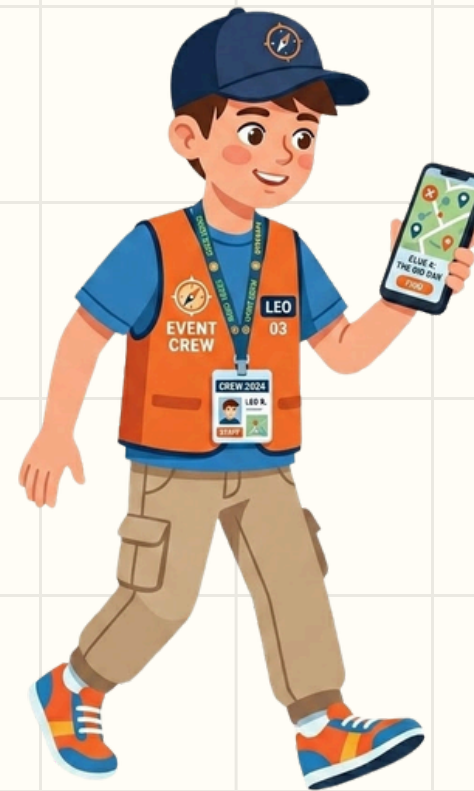
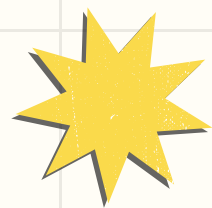




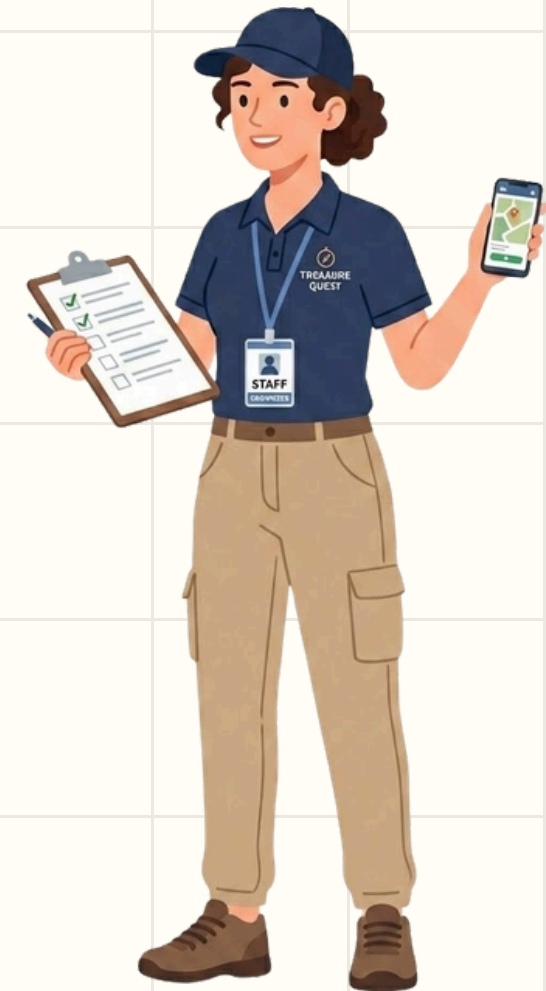
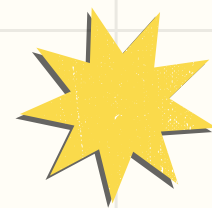
Users



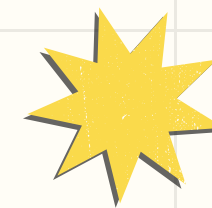
Participant



Crew/Marshall



Event Organizer



Literature Review

Investigation: Understanding Cryptographically Secure
Geofencing & Replay Protection



- ★ Cryptographically: Use of mathematical algorithms to protect location check-ins by ensuring data integrity, authenticity, and message verification.
- ★ Applied: Used to generate dynamic, HMAC-signed QR codes, ensuring checkpoint credentials cannot be forged, screenshotted, or shared by unauthorized participants.
- ★ Secure Geofencing: Coordinates are validated dynamically and access is restricted only to teams physically present within boundaries (Seneviratne et al., 2017).
- ★ Replay Protection: Designed to prevent participants from reusing expired QR codes or intercepted transaction tokens to record fraudulent check-ins (OWASP Foundation, 2023).

Literature Review

Investigation: Implementation Areas & Stakeholders



- ★ Target Organization: Developed and deployed for Dapo Awoknyee Resources (KL Event Crew)
- ★ Implementation Platform: Participant mobile app, crew/marshal verification view, and administrative web console.
- ★ Stakeholders:
 - Event Administrators/Planners
 - Event Participants
 - Field Crews

Literature Review

Investigation: Importance of Cryptographically Secure
Location Verification



- ★ Prevention: Can block coordinates spoofing, static QR duplication, and out-of-bounds checkpoint scans.
- ★ Transaction Authenticity: The cryptographic verification process ensures that each checkpoint completion represents a unique, authenticated transaction, eliminating cheating.
- ★ Credential Protection: Cryptographic storage helps protect sensitive user tokens in the hardware-backed keystore/keychain (Meng et al., 2018).
- ★ Network Resilience: Secure offline synchronization supports continuous event execution even during network dropouts in remote areas (Pothineni, 2024).

Related Works



Turf



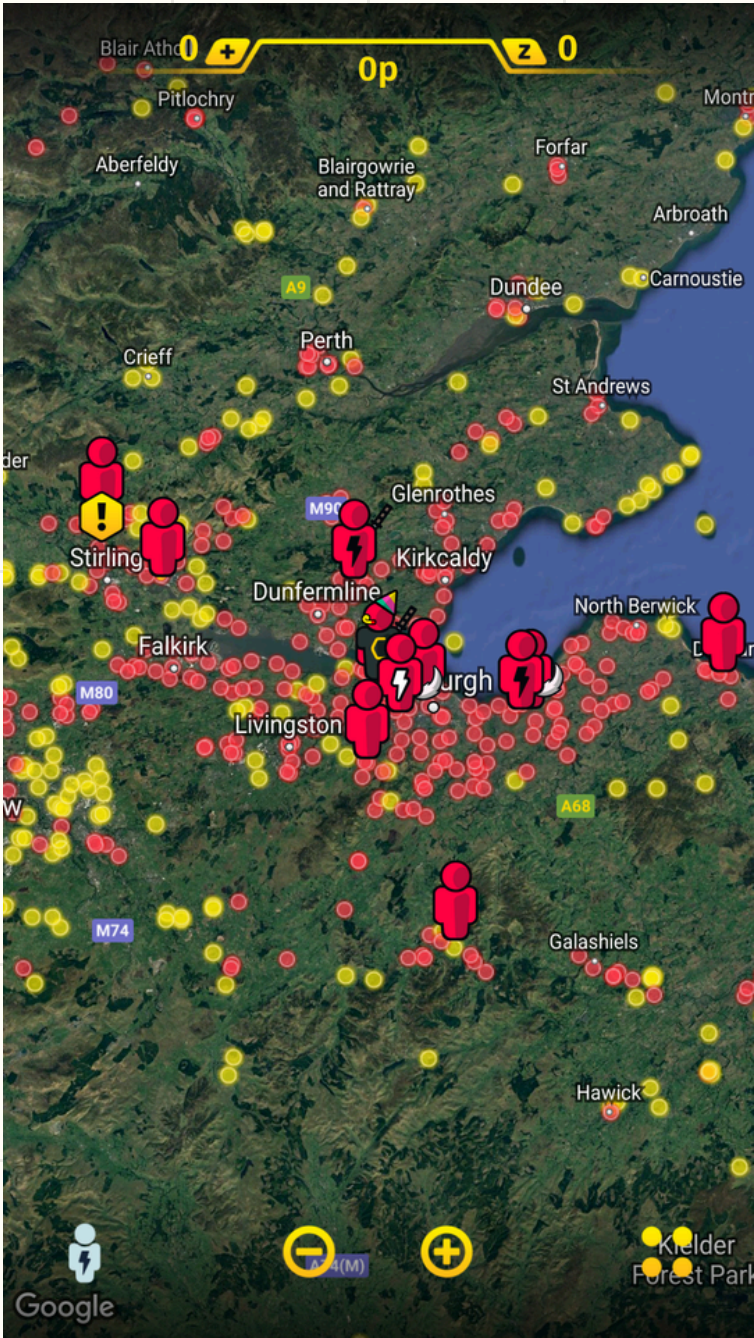
Munzee



Actionbound

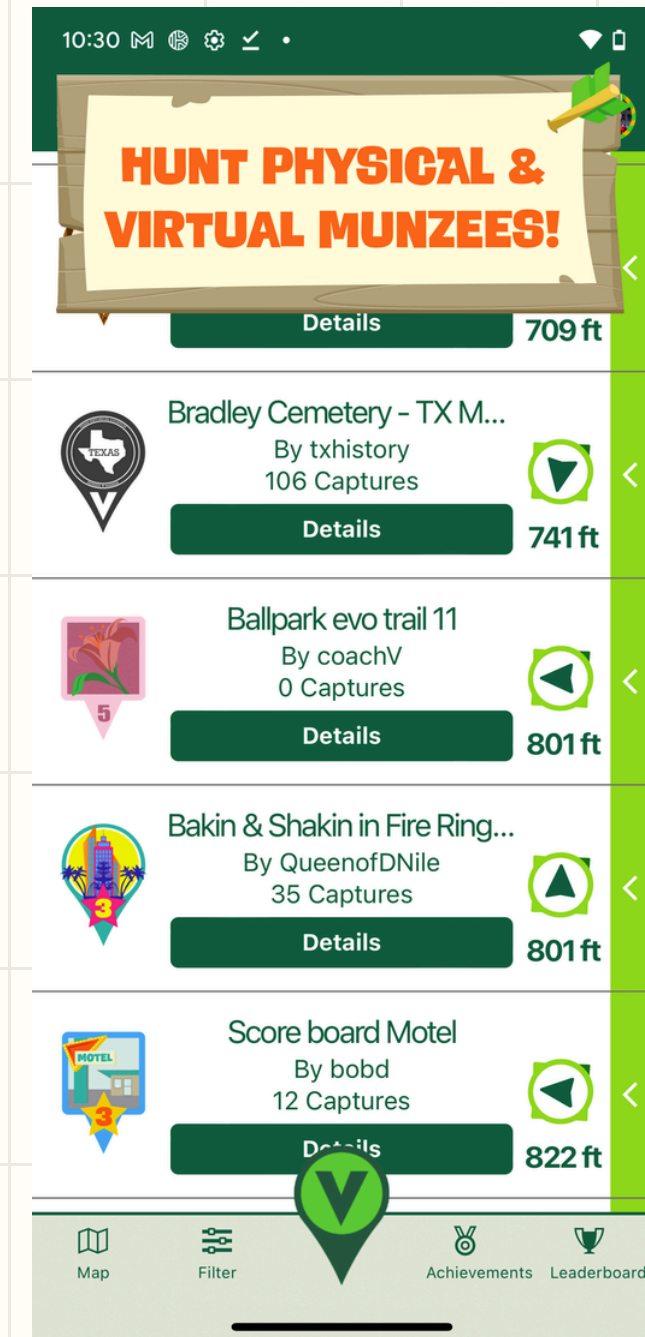
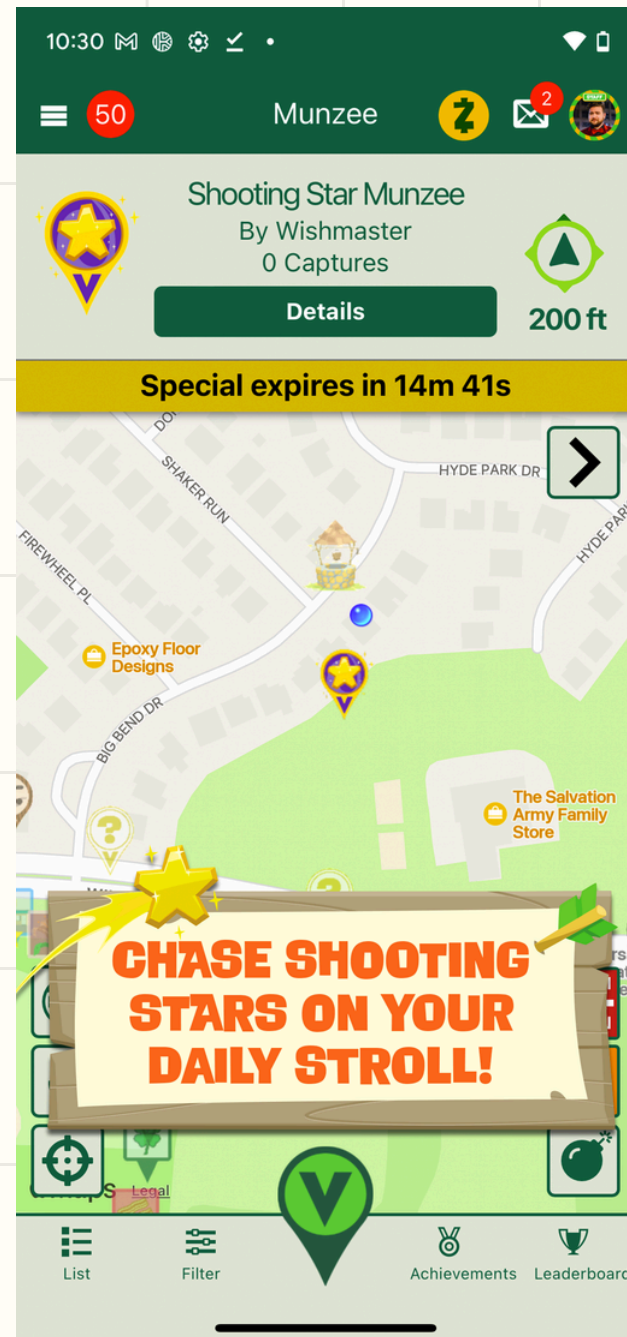
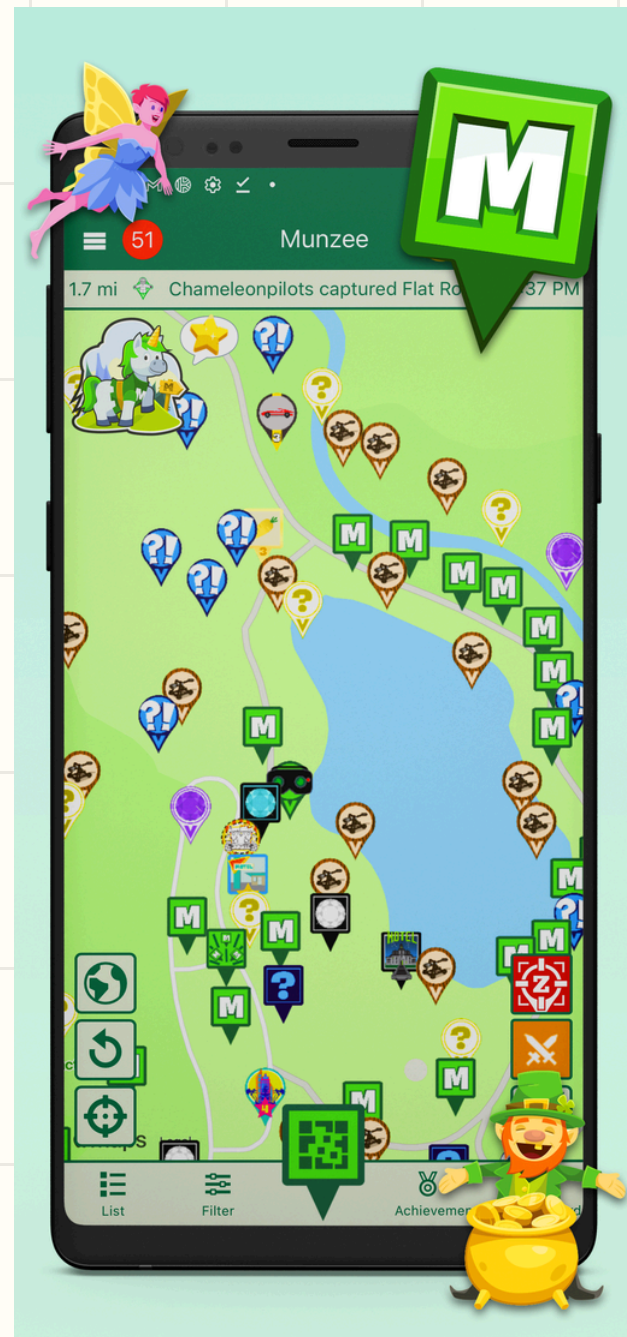


Turf



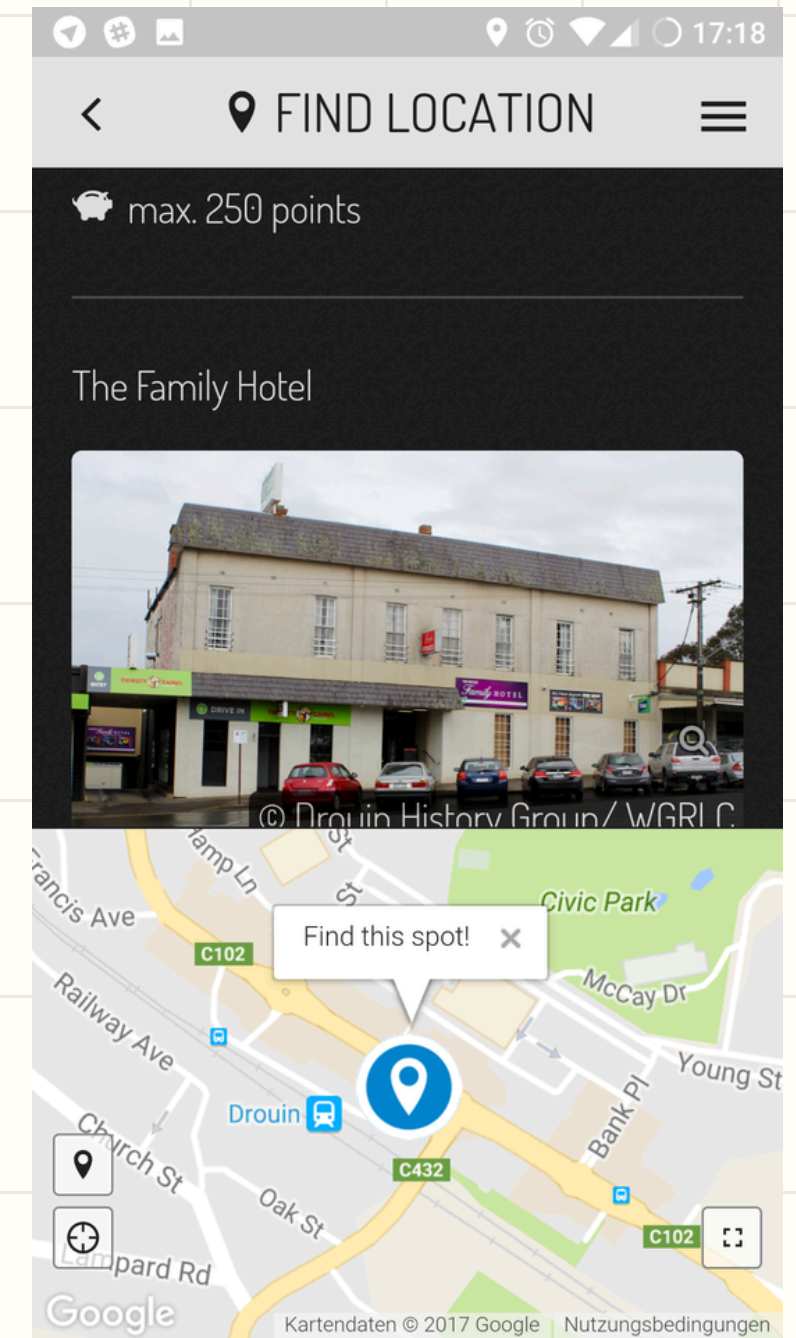
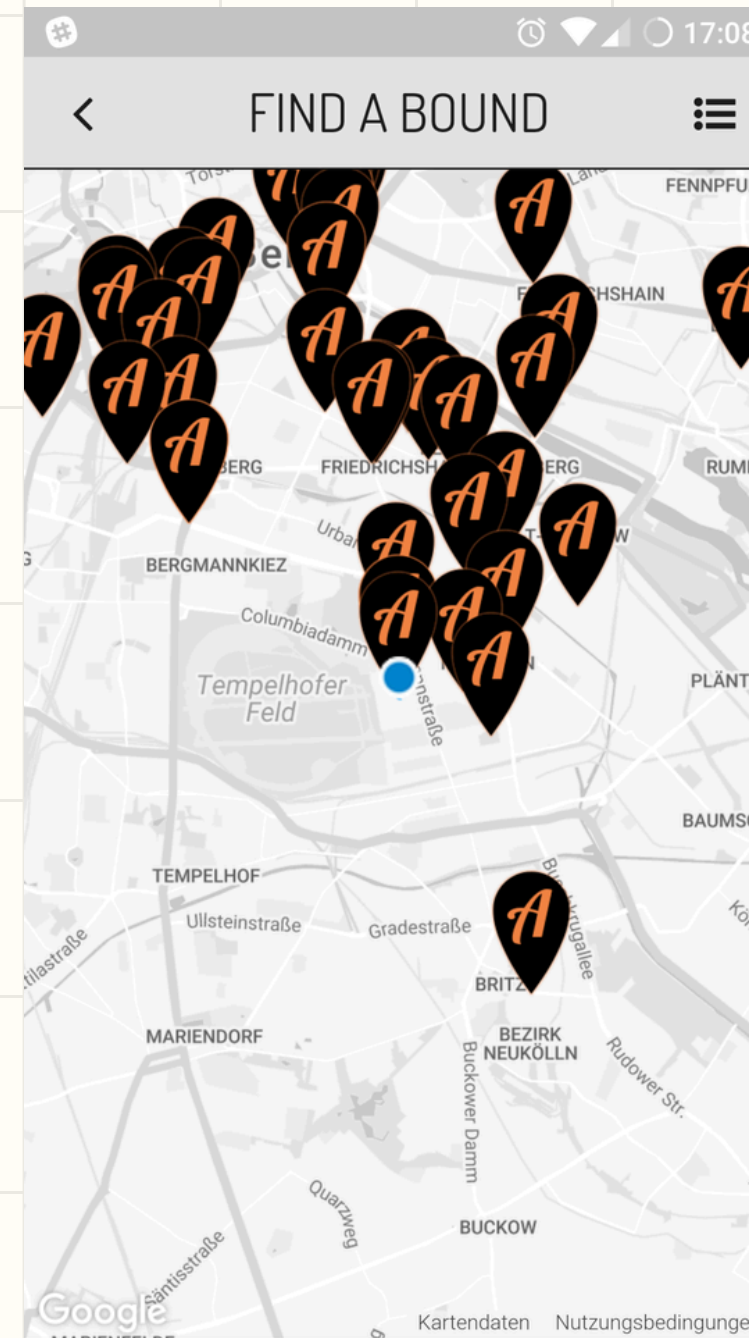
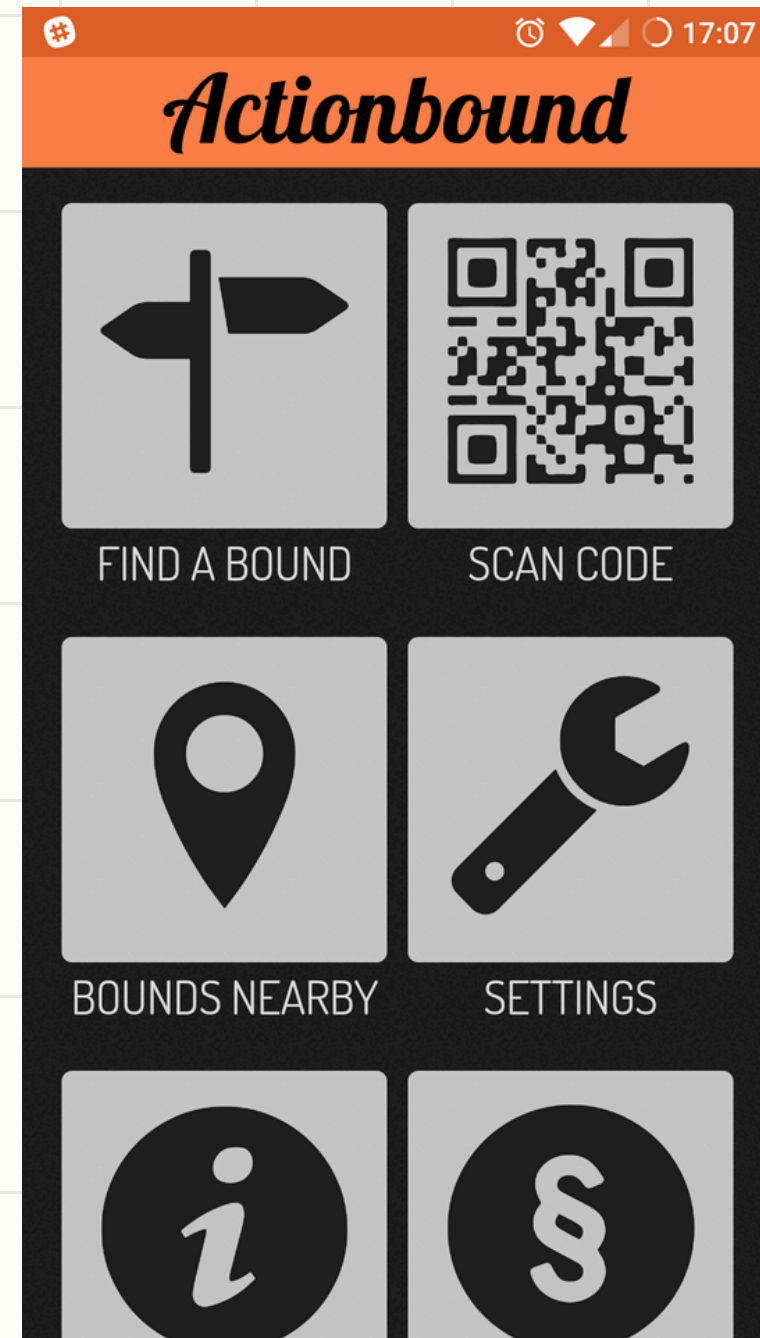


Munzee





Actionbound

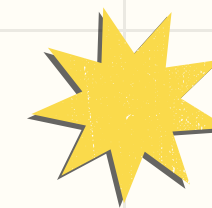


Criteria / Project	Turf	Munzee	Actionbound	XploreQuest
Features	Territory capture, monthly rounds, global leaderboards, virtual zones.	Scavenger hunting, virtual/physical pins, in-game item store, clan battles.	Custom quest creator, offline bounds, quizzes, multimedia missions.	Secure dynamic QR scanning, offline queue syncing, real-time marshal dashboard.
User	Individual fitness gamers.	Individual collectors and casual explorers.	Educators, corporate groups, and casual tourists.	Event Participants, Event Marshals, and Event Organizers (KL Event Crew).
Navigation/ Interface	Real-time map with active player/zone markers.	Map with pins (physical vs. virtual markers).	Custom maps, list of quests, and directional compass.	Interactive map (Leaflet/Google Maps) with responsive web portal for marshals.
Pickup Verification	Passive GPS proximity (staying in a zone for ~30 seconds).	Scanning static QR code stickers OR passive GPS range checks.	Scanning static QR codes, GPS coordinates, or uploading media proofs.	Dynamic QR code scanning (time-sensitive signature) + Server-side GPS verification.
Software	Android & iOS mobile apps.	Android & iOS mobile apps.	Android & iOS mobile apps, web-based creator portal.	Cross-platform Expo (React Native) mobile app, web dashboard, and Firebase backend.
Hardware	Smartphone (GPS, internet connection).	Smartphone (GPS, camera, internet connection).	Smartphone (GPS, camera, local storage, microphone).	Smartphone (GPS, camera, SecureStore, local SQLite database for offline queue)
Security Elements	Basic server validation (susceptible to GPS spoofing).	Basic location validation (highly vulnerable to QR sharing/screenshots).	Basic GPS/QR match (vulnerable to QR screenshot sharing).	Multi-layer security: Dynamic HMAC-SHA256 signatures, GPS velocity telemetry checks, Expo SecureStore.
Result	Fitness tracking and competitive zone control.	Casual point collection and global player ranking.	Educational tours, team building, and custom gamified events.	Secured checkpoint validation preventing spoofing, screenshot sharing, and event cheating.

Agile Methodology



- Flexibility & Iteration
- Adaptability
- Continuous Testing & Feedback



Plan

Construct the product backlog.

Outline the security.

Write secure backend APIs and validation checks.

Conduct security tests.

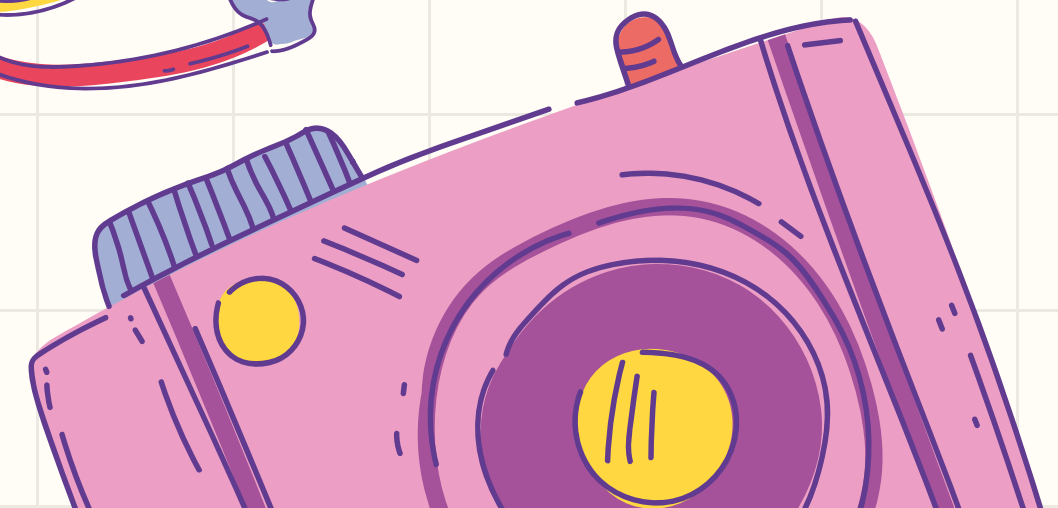
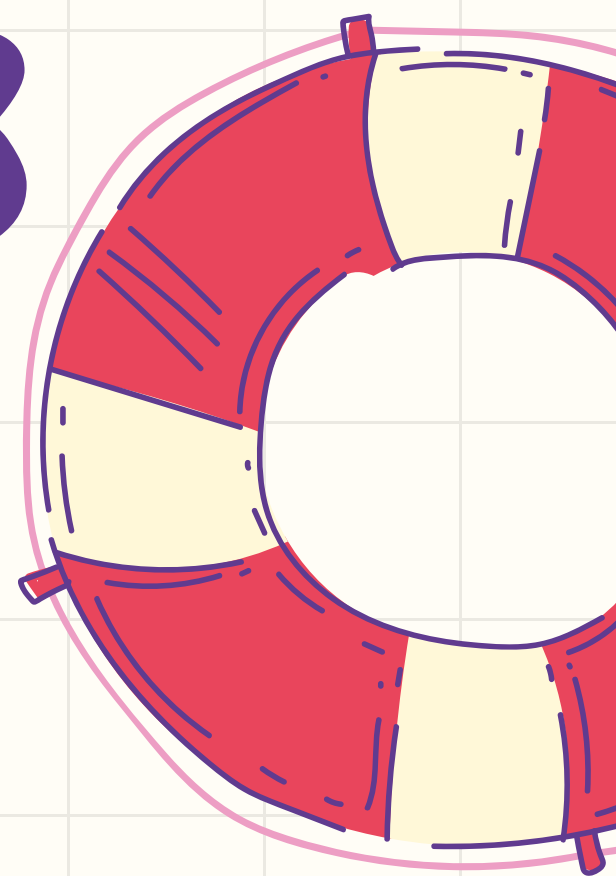
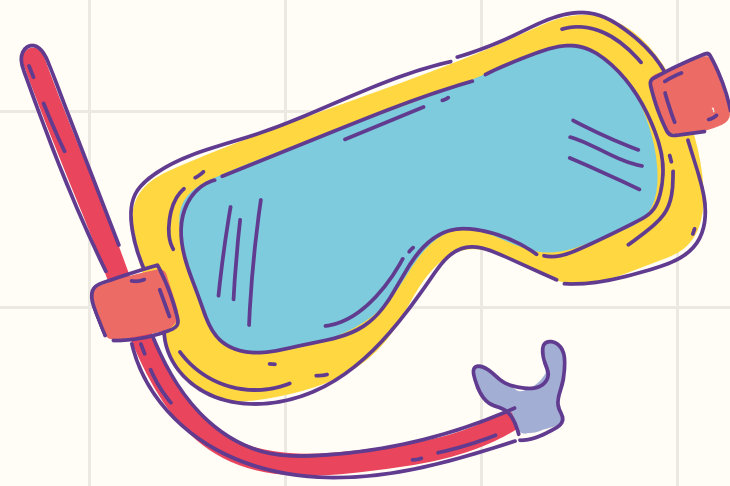
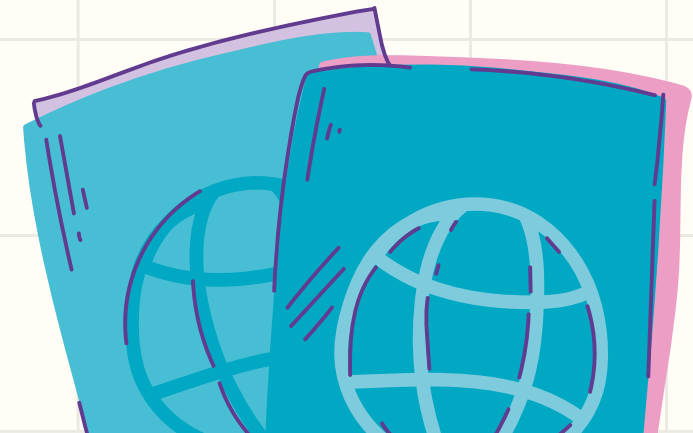
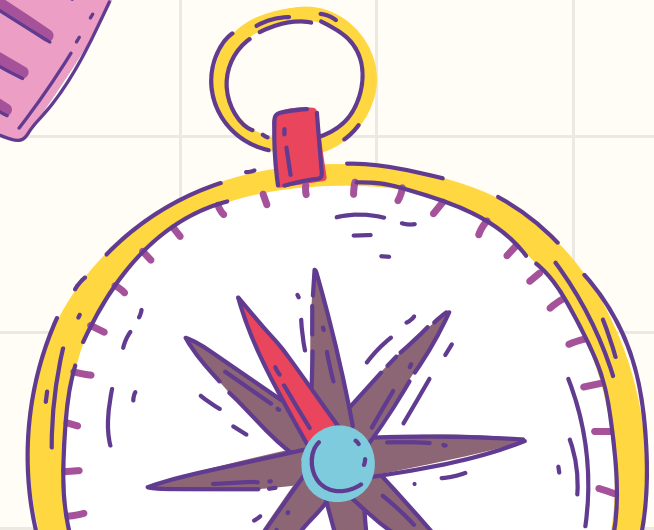
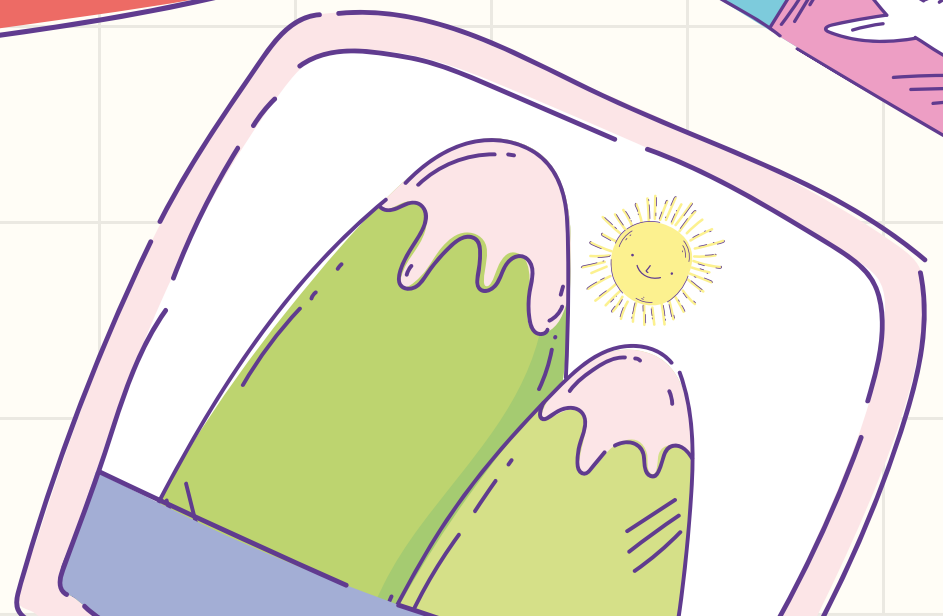
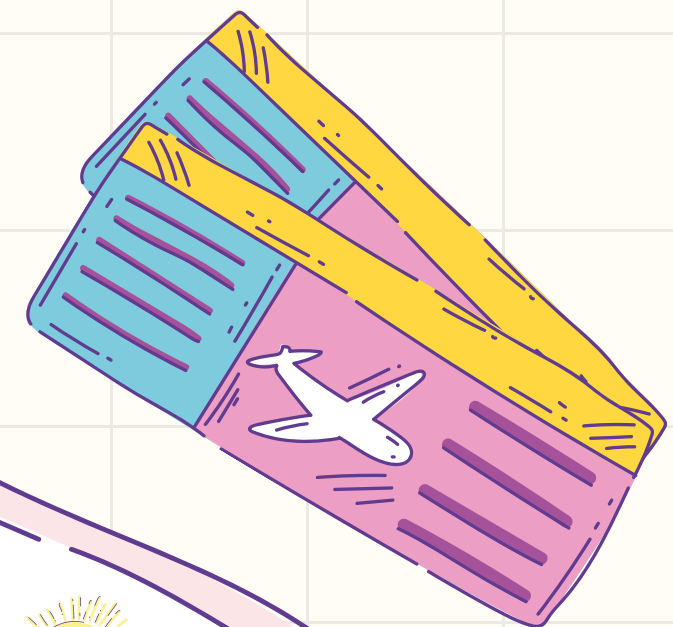
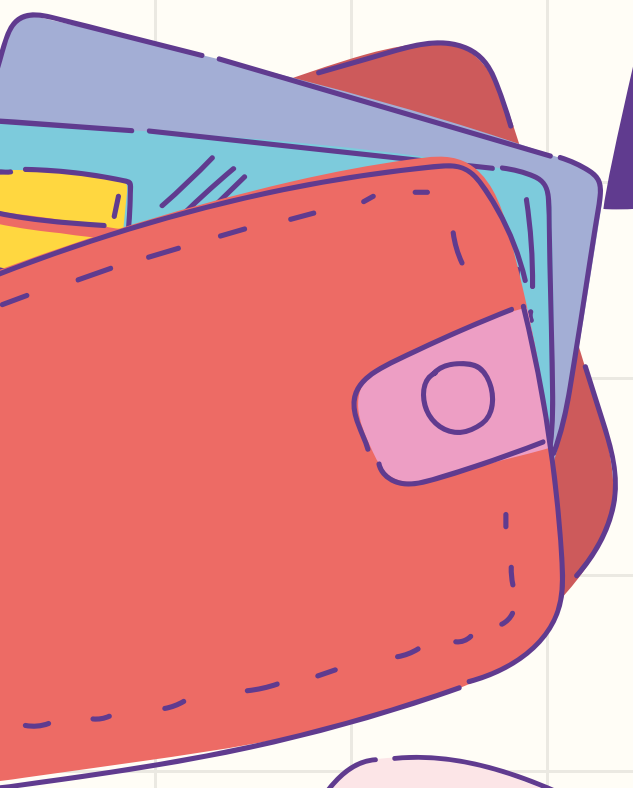
Deploy the web dashboard

Check sync latency,
analyze logs, and patch
post-deployment bugs.

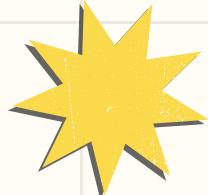
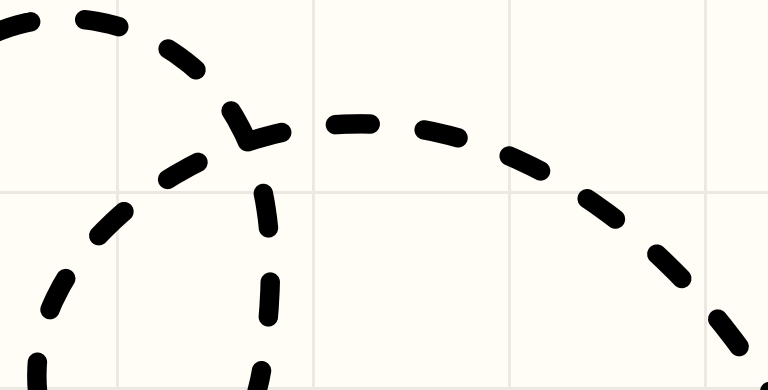
Scrum Framework

Sprint-Based Iteration	Security Backlog Prioritization	Continuous Risk Reduction
Iterative 2-Week Sprints	Early Threat Countermeasures	Isolated Testing Gates
Modular development of the Participant App, Marshal View, and Organizer Dashboard.	Map and prioritize critical security rules (HMAC signatures & GPS velocity checks) in initial sprints.	Run automated Jest tests at the end of each sprint to prevent late-stage system integration failures.

Requirements



DATA GATHERING TECHNIQUES

- An online consultation was conducted with Puan Nurul Husna Binti Mustafa.
 - Purpose: To collect detailed, practical information about the crew's current event coordination workflows and checkpoint validation challenges.
 - Interview: Essential in shaping the core features of XploreQuest (e.g., skip checkpoints, dynamic QR verification, and manual crew overrides).
- 
- Distributed to university students, youth, and the general public who participate in outdoor exploration events.
 - Targeted a minimum of 50 respondents to collect a statistically useful sample.
 - Purpose: To analyze users' experiences with network drops, preferences for offline-capable apps, and perceptions of location-based cheating (GPS spoofing).
- 

Hardware Requirement



Specification	Description
Model	HONOR MagicBook X 15 (BBR-WAX9)
Processor	Intel® Core™ i3-10110U CPU @ 2.10GHz
RAM	8.00 GB
Storage	256 GB SSD
Graphic Card	Intel® UHD Graphics
System Type	64-bit Operating System, x64-based Processor
Windows Edition	Windows 10 Home Single Language

Software Requirement

Software Category	Tool / Platform
Operating System	Windows 10 Home
Code Editor (IDE)	Visual Studio Code (VS Code)
Mobile Framework	Expo (React Native)
Programming Language	JavaScript / TypeScript
Backend & Database	Google Firebase

Backend & Database	Google Firebase
Version Control	Git & GitHub
Testing / Emulation	Android Studio (Emulator) & Expo Go
UI/UX Design Tool	Figma
Database Management	Firebase Console
Documentation & Slides	MS Office / Google Docs

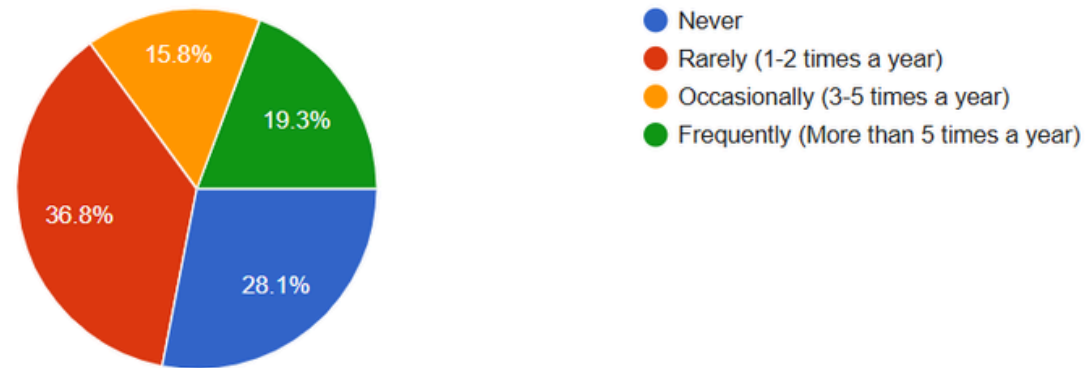
Questionnaire Analysis



Q4. How often do you participate in outdoor adventure races, scavenger hunts, or exploratory events (e.g., Explorace, running events, treasure hunts)?

[Copy chart](#)

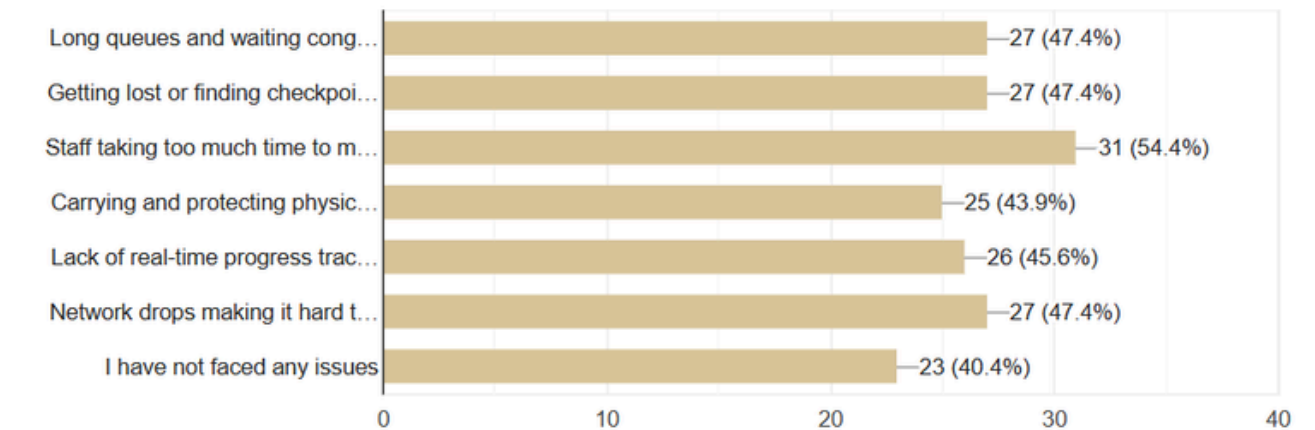
57 responses



Q7. What are the main difficulties or bottlenecks you have encountered during checkpoint-based events? (Select all that apply)

[Copy chart](#)

57 responses

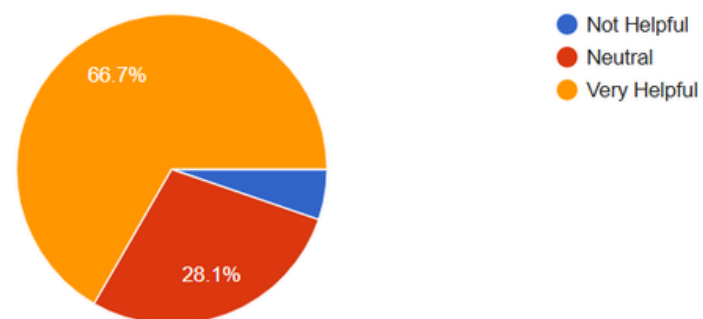


Section 4: Feasibility of Proposed Features (XploreQuest)

Q12. Rate how helpful the following features would be during an outdoor adventure event:

[Copy chart](#)

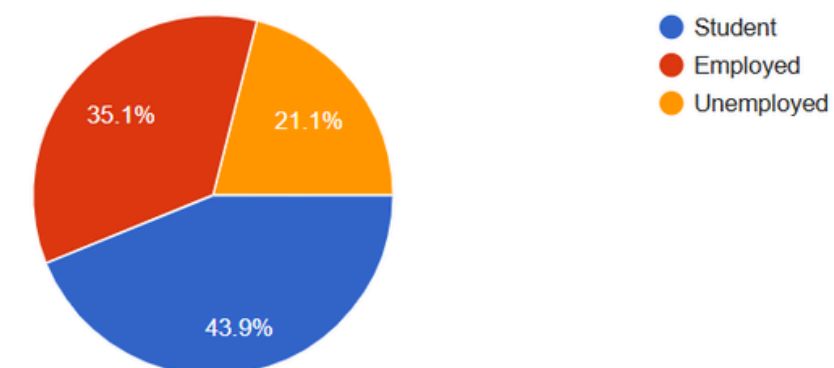
57 responses



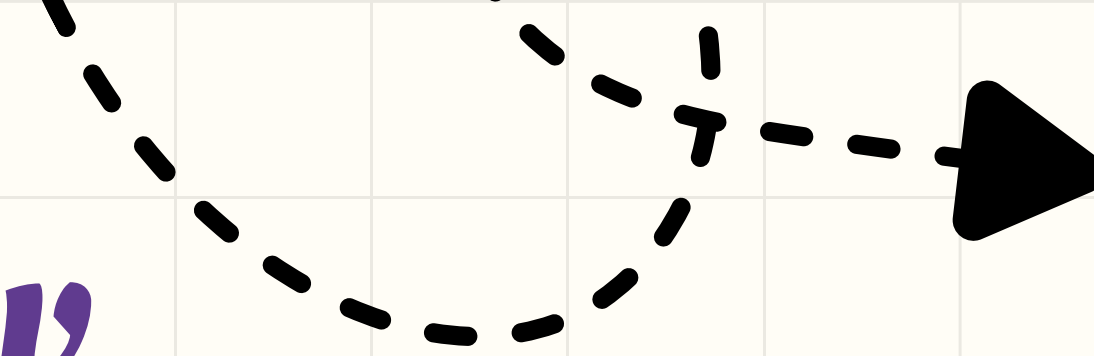
Q3. Occupation / Status

[Copy chart](#)

57 responses

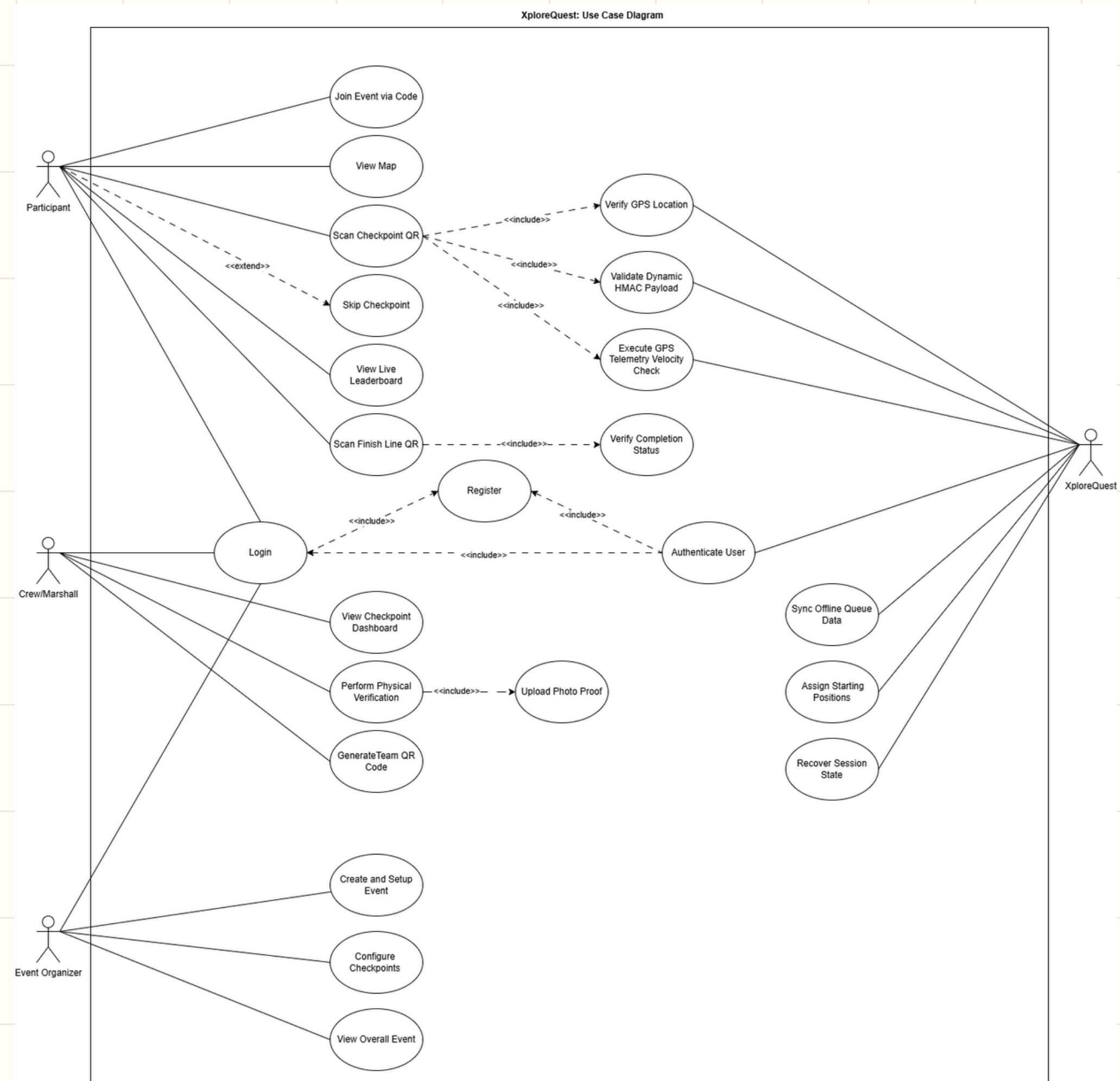
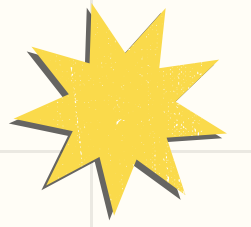


Interview Analysis



- Bottleneck Prevention (Skip Logic): Allowed teams to temporarily skip congested checkpoints, requiring them to complete all tasks before unlocking the finish line.
- Automated Timers & Penalties: Implemented countdown timers for tasks and automatic time penalties (e.g., +5 minutes) to ensure fair time tracking.
- Real-Time Progress Indicators: Added progress bars, completion checkmarks (✓), and warning pop-ups to notify players of missing checkpoints.

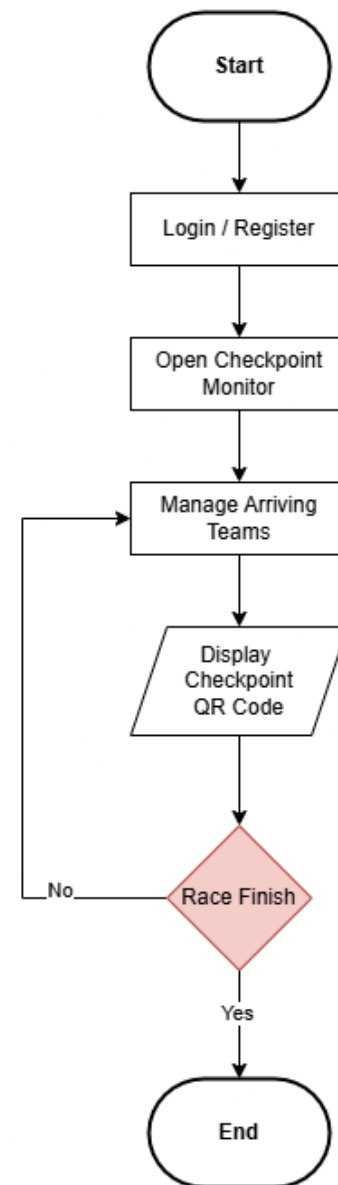
Use Case Diagram



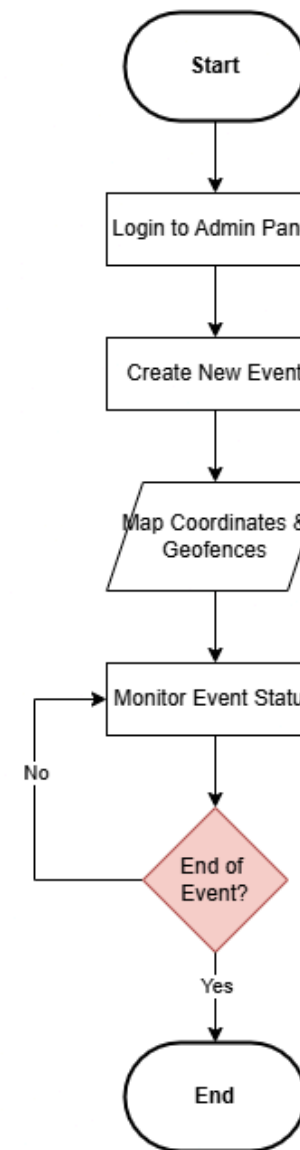
Flowchart Diagram



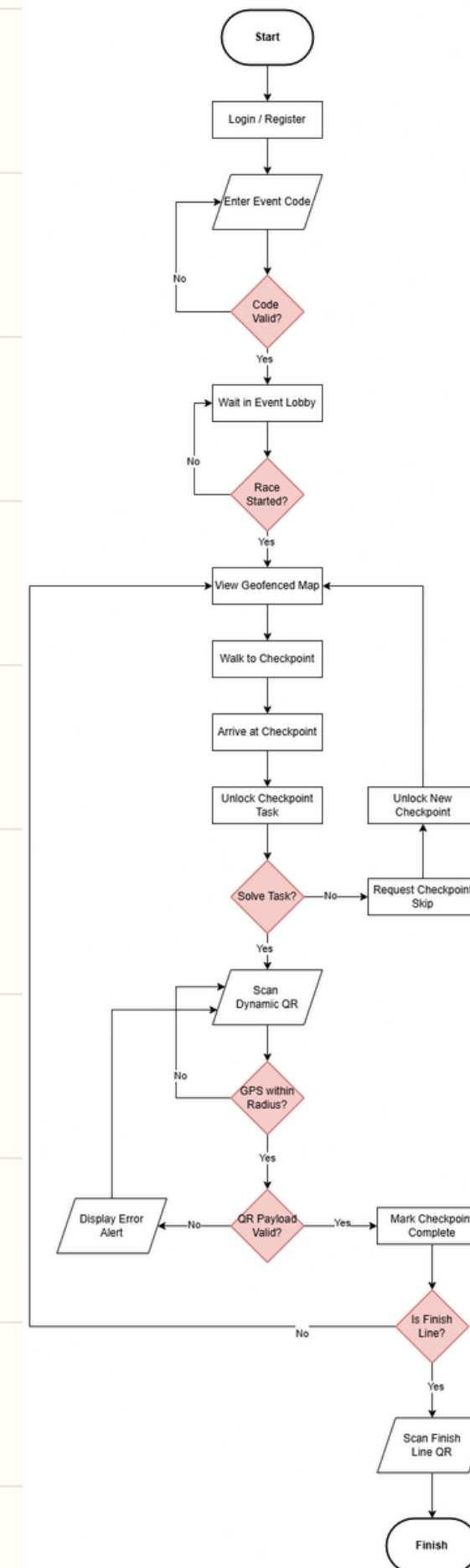
Crew/Marshall Workflow Flowchart



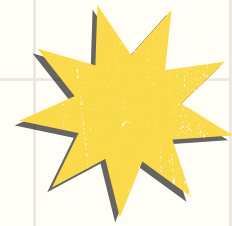
Event Organizer Flowchart



Participant Workflow Flowchart



Conclusion



In conclusion, the XploreQuest project aims to develop a secure mobile adventure application that prevents cheating, validates checkpoints through geofencing and dynamic QR codes, and ensures offline capability. These objectives focus on solving manual verification bottlenecks and GPS location spoofing. I hope this project establishes a solid foundation for future development in Final Year Project 2. I believe XploreQuest can improve event security, operational efficiency, and trust for outdoor races.

